

Substitution into Expressions and Formulae

Year 10 Algebra

What is substitution? Replacing a letter (a variable) with a number, then working out the value of the expression. It's the bridge between algebra and real-world formulae — especially the ones you'll use in finance.

The 3-step method: **Identify** what's given and what to find. $\Rightarrow$ **Apply** by replacing each variable with its value. $\Rightarrow$ **Verify** that the answer makes sense.

Worked Example 1 — Gross Pay

Your pay P (\$) for working h hours is given by $P = 25h$. Find P when $h = 8$.

Identify: Given: $h = 8$. Find: P .

Apply: $P = 25 \times 8 = \$200$

Verify: At \$25/hour, 8 hours = $8 \times 25 = \$200$. ✓

Worked Example 2 — Simple Interest

The interest I earned on a savings account is $I = \frac{PRT}{100}$, where P = principal (\$), R = rate (%), T = years.

Find the interest on \$400 invested at 5% for 3 years.

Identify: Given: $P = 400$, $R = 5$, $T = 3$. Find: I .

Apply: $I = \frac{400 \times 5 \times 3}{100} = \frac{6000}{100} = \60

Verify: 5% of \$400 is \$20/year. Over 3 years: $3 \times 20 = \$60$. ✓

Now you try. Work through Foundation first. Move to Standard when you're confident. Higher is a stretch.

Foundation

F1. Find the value of $3x + 5$ when $x = 4$.

F2. Find the value of $2y - 3$ when $y = 8$.

F3. Pay is given by $P = 22h$. Find P when $h = 7$ hours.

Standard

S1. A car hire company charges $C = 50 + 35d$ dollars, where d is the number of days. Find the cost for 4 days.

S2. Find $4m + 3n$ when $m = -2$ and $n = 5$.

S3. Use $I = \frac{PRT}{100}$ to find the interest when $P = \$800$, $R = 6$, $T = 4$.

Higher

H1. Find $2a^2 - 3ab$ when $a = -1$ and $b = 4$.

H2. Use $A = P(1 + r)^n$ to find A when $P = \$2000$, $r = 0.04$, $n = 3$. Round to the nearest cent.

Exit ticket. Find $5x - 2$ when $x = 3$. Answer = _____